

Dimensional family

$$A_{r,k} = S\tau^k \rho^{-r}$$

weights,
domain

Synthesized scalar law

$$P(D)y = N(D)u$$

elimination

initial data,
constraints

Declared states and ports

$$\dot{x} = Fx + Bu$$

state image

Admissible observation

$$y = Cx, \text{ compatible jets}$$

Effective derivative order, observable order and physical state dimension are separate quantities.